Optimization of Emergency Supply Distribution Networks using Mixed Integer Linear Programming

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Abstract

Efficient logistics planning is critical for maintaining reliable supply distribution in large campuses. This project develops an optimization model for emergency supply distribution using Mixed Integer Linear Programming (MILP). The model determines which warehouses should be operational and how supplies should be distributed to campus facilities while minimizing the total annual cost.

The optimization considers construction costs, operational costs, transportation costs, and warehouse capacity constraints. A redundancy constraint ensures that exactly two warehouses remain active for system reliability. The PuLP optimization library is used to solve the MILP model, and Folium is used to visualize the resulting distribution network on an interactive geographic map.

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1 Introduction

Efficient supply chain management plays an essential role in ensuring that critical resources are distributed reliably and cost-effectively. Large campuses often require centralized storage facilities to distribute supplies to multiple buildings such as medical centers, dormitories, and libraries.

Optimization techniques provide a systematic way to determine the most efficient logistics configuration. Linear programming and Mixed Integer Linear Programming are commonly used to model transportation and facility location problems.

This project develops an optimized emergency supply distribution network for Campus City using MILP. The model determines the optimal warehouse configuration and shipping routes while minimizing operational costs.

2 Problem Statement

The objective of this project is to design a cost-efficient supply distribution network for Campus City.

The system must:

- Determine which warehouses should be operational
- Ensure all facilities receive their required supply
- Minimize total annual logistics cost
- Respect warehouse capacity limits
- Maintain redundancy by activating exactly two warehouses

3 Mathematical Formulation

3.1 Decision Variables

- y_i : Binary variable indicating whether warehouse i is active
- x_{ij} : Units transported from warehouse i to facility j

3.2 Objective Function

$$Z = \sum_{i \in W} \left(\frac{C_i}{10} + 365 O_i \right) y_i + \sum_{i \in W} \sum_{j \in F} (4.35 \times x_{ij})$$

3.3 Constraints

Demand Satisfaction

$$\sum_{i \in W} x_{ij} = D_j \times 365$$

Warehouse Capacity

$$\sum_{j \in F} x_{ij} \leq S_i \times 365 \times y_i$$

Redundancy Requirement

$$\sum_{i \in W} y_i = 2$$

4 Dataset

4.1 Facility Demand Dataset

Facility	Daily Demand	Annual Demand
MED_CENTER	80	29200
ENG_BUILDING	30	10950
SCIENCE_HALL	35	12775
DORM_A	55	20075
DORM_B	45	16425
LIBRARY	25	9125

Table 1: Facility Demand Dataset

4.2 Warehouse Dataset

Warehouse	Capacity	Construction Cost	Daily Operation Cost
WH_NORTH	400	300000	800
WH_SOUTH	350	280000	700
WH_EAST	450	320000	900

Table 2: Warehouse Dataset

4.3 Warehouse Coordinates

Warehouse	Latitude	Longitude
WH_NORTH	10.130	76.465
WH_SOUTH	10.120	76.450
WH_EAST	10.125	76.470

Table 3: Warehouse Coordinate Dataset

5 Methodology

The optimization process follows these steps:

1. Define facility demand and warehouse parameters

2. Construct MILP optimization model
3. Define objective function and constraints
4. Solve the model using PuLP CBC solver
5. Extract optimal warehouse selection
6. Visualize results using Folium mapping

6 Results

The optimization model selects the following warehouses:

- WH_NORTH
- WH_SOUTH

Total Annual Cost:

$$\$1,034,192.50$$

Budget Limit:

$$\$1,500,000$$

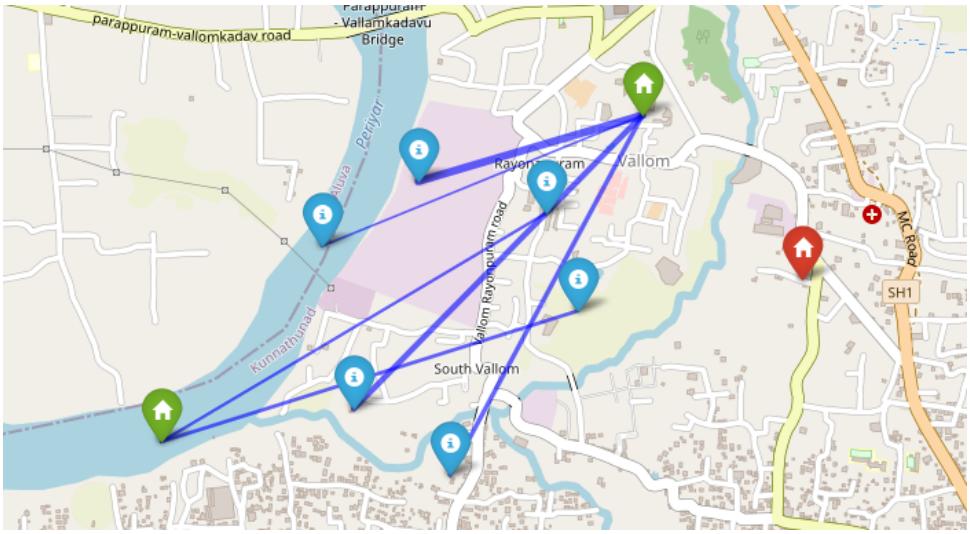

Figure 1: Optimized Distribution Network

6.1 Project Repository

The complete implementation and project files are available on GitHub:
https://github.com/athira-anil2327/Campus-Logistic-Optimization-problem

7 Code Snippet

```
from IPython.display import display

# ----------------------------
# 9. FOLIUM VISUALIZATION
# ----------------------------

m = folium.Map(location=[10.125, 76.460], zoom_start=15)

# Warehouses
for w, coords in warehouse_locations.items():

    is_open = lp.value(y[w]) == 1
    color = "green" if is_open else "red"

    folium.Marker(
        location=coords,
        popup=f"{w} | {'OPEN' if is_open else 'CLOSED'}",
        icon=folium.Icon(color=color, icon="home")
    ).add_to(m)

# Facilities
for f, coords in facility_locations.items():

    folium.Marker(
        location=coords,
        popup=f,
        icon=folium.Icon(color="blue", icon="info-sign")
    ).add_to(m)

# Routes
for w in warehouses:

    if lp.value(y[w]) == 1:

        for f in facilities:

            shipment = lp.value(x[w,f])

            if shipment > 0:

                folium.PolyLine(
                    locations=[warehouse_locations[w],
                       facility_locations[f]],
                    weight=shipment/5000,
                    color="blue",
                    opacity=0.6
                ).add_to(m)

# Display map in output
display(m)
```

8 Conclusion

This project demonstrates how optimization techniques can be applied to logistics planning. Using Mixed Integer Linear Programming, the system identifies the most cost-efficient warehouse configuration while maintaining operational reliability.

The results show that activating two warehouses (North and South) satisfies all facility demands while staying well within the budget.

9 References

1. PuLP Linear Programming Documentation
2. Python Optimization Libraries
3. Logistics Optimization Research Papers